

María Alejandra Schild Zerene

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Studies

Pontificia Universidad Católica de Chile (UC Chile) <i>PhD in Engineering Sciences, specializing in Computer Science</i>	Santiago, Chile 2025-present
Pontificia Universidad Católica de Chile (UC Chile) <i>Master's in Mathematics</i> ○ GPA: 6.6/7.0 ○ Thesis title: <i>Computational Complexity of Counting Shellings of a Simple Graph</i>	Santiago, Chile 2022-2024
Pontificia Universidad Católica de Chile (UC Chile) <i>Bachelor's in Mathematics</i> ○ GPA: 6.82/7.0	Santiago, Chile 2018-2021

Awards and Scholarships

National Agency for Research and Development (ANID) <i>ANID National Doctoral Scholarship</i> Ranked 10th out of 3359 applications	Chile 2025
Chilean Mathematical Society (SOMACHI) <i>Honorable Mention, Jorge Billeke Award</i> Academic excellence among graduates of Mathematics bachelor's or Mathematical Engineering programs across Chile	Chile 2022
National Agency for Research and Development (ANID) <i>ANID National Master's Scholarship</i> Ranked 6th out of 1895 applications	Chile 2022
Pontificia Universidad Católica de Chile (UC Chile) <i>Academic Excellence Award</i>	Santiago, Chile 2021
Faculty of Mathematics, UC Chile <i>Rolando Chuaqui Scholarship</i> Award for academic excellence	Santiago, Chile 2020
International Mathematical Olympiad (IMO) <i>Bronze Medal</i>	Rio de Janeiro, Brazil 2017

Teaching

Escuela Militar del Libertador Bernardo O'Higgins <i>Introduction to Programming</i> for second-year cadets, under an academic agreement with UC Chile	Santiago, Chile 1st semester 2026
Faculty of Mathematics, UC Chile <i>Introduction to Geometry</i> for Mathematics undergraduates and secondary mathematics teacher trainees	Santiago, Chile 2nd semester 2024
Faculty of Mathematics, UC Chile <i>Introduction to Calculus</i> for students in the Bachelor in Computer Science program	Santiago, Chile 1st semester 2024
Faculty of Mathematics, UC Chile <i>Instructor in the Mathematical Reasoning Workshop (TRM) program</i>	Santiago, Chile April-November 2022

Teaching Assistantships

Master's in Artificial Intelligence, UC Chile

Neurosymbolic Artificial Intelligence (August–September 2026).

Department of Computer Science, UC Chile

Topics in Computer Science: communication complexity (2025-II); Logic for Computer Science (2025-I).

Faculty of Mathematics, UC Chile

Foundations of Mathematics (2024-I); Introduction to Combinatorics (2023-I); Calculus I (2022-I); Mathematics Help Center (2020-II, 2021-I); Mathematics Workshop (2019-I, 2020-I).

Talks

Computational Complexity Seminar (Department of Computer Science, UC Chile)

Search-to-decision reducibility for NP problems (1 session)
Computational complexity of the Tutte polynomial (2 sessions)

Santiago, Chile

April 2026

September 2025

Computational Complexity Seminar (Faculty of Mathematics, UC Chile)

Connections between algebraic combinatorics and the complexity class $\#P$ (8 sessions)

Santiago, Chile

August–November 2023

Algebraic Combinatorics Seminar (Faculty of Mathematics, UC Chile)

Shelling orders and Stanley-Reisner rings (3 sessions)
Deletion-contraction and the Tutte polynomial of matroids (2 sessions)

Santiago, Chile

May–June 2023

September 2022

Mathematics Club (Faculty of Mathematics, UC Chile)

The Kruskal-Katona theorem for simplicial complexes (1 session)

Santiago, Chile

June 2022

Conferences and Schools Attended

Schloss Dagstuhl - Leibniz Center for Informatics

Knowledge Compilation in Artificial Intelligence, Databases, and Formal Methods

Wadern, Germany

May 26–29, 2026

Max Planck Institute for Mathematics in the Sciences

Summer School in Algebraic Combinatorics

Leipzig, Germany

July 15–19, 2024

Encuentro Colombiano de Combinatoria (ECCO)

Bridges between Algebra and Combinatorics

Popayán, Colombia

June 17–21, 2024

CIMPA Research School

Crossroads of Geometry, Representation Theory and Higher Structures

Puerto Madryn, Argentina

March 13–24, 2023

Languages

- Spanish: Native
- English: Fluent
- German: Intermediate

Other Activities

Pan-American Girls' Mathematical Olympiad (PAGMO)

Coordinator and member of the Problem Selection Committee

Online

2021 and 2022